Phase 1: Modelling, Equilibria and Linearization of the Acrobot

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Abstract. We verify the given Acrobot model, find its zero-torque equilibria, linearize at each, and derive the transfer functions from the elbow torque τ to both joint angles. Phase 2 (stability analysis) starts on page 4 and Phase 3 (control) on page 8.

I. Nonlinear Model

A. System and Parameters. Two-link pendulum (Fig. 1) with a passive shoulder and a motor at the elbow [1], [2].

- Angles: q_1 = link 1 from the downward vertical; q_2 = link 2 relative to link 1; $\varphi = q_1 + q_2$.
- State: $x = [q_1,\ q_2,\ \dot{q}_1,\ \dot{q}_2]^T$.
- Input: elbow motor torque τ, $|\tau| \le 20\,\mathrm{N\,m}$, the only quantity we command ($\tau = 0$ in this phase; set by a controller in later phases).
- $m_1 = m_2 = 1$ kg, $l_1 = 1$ m, $l_2 = 2$ m, $l_{c1} = 0.5$ m, $l_{c2} = 1$ m, $I_1 = 0.083$, $I_2 = 0.33\ \mathrm{kg\,m^2}$, $g = 9.81\ \mathrm{m/s^2}$.

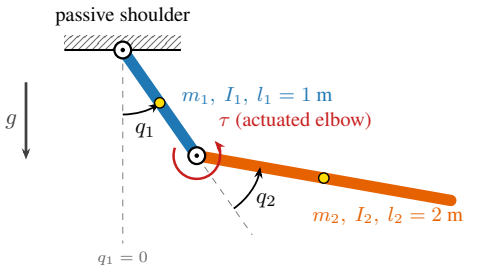


Fig. 1. Acrobot schematic (to scale); dots mark the centres of mass.

B. Given Equations of Motion.

$$M(q)\ddot{q} + C(q,\dot{q})\dot{q} = \tau_g(q) + B\tau, \qquad B = [0,\ 1]^T \tag{1}$$

(B applies τ to the elbow equation only; $c_2 = \cos q_2$, $s_2 = \sin q_2$.)

C. Derivation. *1) Potential energy and gravity torque:*

$$U = -(m_1 l_{c1} + m_2 l_1) g \cos q_1 - m_2 l_{c2} g \cos\varphi = -14.715\cos q_1 - 9.81\cos\varphi, \qquad \tau_g = -\frac{\partial U}{\partial q} = -9.81\begin{bmatrix} 1.5\sin q_1 + \sin\varphi \\ \sin\varphi \end{bmatrix}. \tag{2}$$

2) Kinetic energy and inertia matrix: the velocity of centre 2 is the elbow velocity ($l_1\dot{q}_1$) plus the rotation about the elbow ($l_{c2}\dot{\varphi}$), at angle q_2 apart (law of cosines):

$$T = \tfrac{1}{2}(m_1 l_{c1}^2 + I_1)\dot{q}_1^2 + \tfrac{1}{2} I_2 \dot{\varphi}^2 + \tfrac{1}{2} m_2 \left(l_1^2\dot{q}_1^2 + l_{c2}^2\dot{\varphi}^2 + 2 l_1 l_{c2} c_2 \dot{q}_1\dot{\varphi}\right) = \tfrac{1}{2}\dot{q}^T M \dot{q}, \quad M(q) = \begin{bmatrix} 2.663 + 2c_2 & 1.33 + c_2 \\ 1.33 + c_2 & 1.33 \end{bmatrix}. \tag{3}$$

3) Coriolis term (Euler–Lagrange): with $\partial T/\partial \dot{q} = M\dot{q}$ and $\dot{M} = \frac{\partial M}{\partial q_2}\dot{q}_2 = -s_2\dot{q}_2 \left[\begin{smallmatrix} 2 & 1 \\ 1 & 0 \end{smallmatrix}\right]$,

$$\frac{d}{dt}\frac{\partial T}{\partial \dot{q}} - \frac{\partial T}{\partial q} = \tau_g + B\tau \ \Rightarrow\ M\ddot{q} + \underbrace{\left(\dot{M}\dot{q} - \frac{\partial T}{\partial q}\right)}_{=C\dot{q}} = \tau_g + B\tau, \qquad C(q,\dot{q}) = s_2\begin{bmatrix} -2\dot{q}_2 & -\dot{q}_2 \\ \dot{q}_1 & 0 \end{bmatrix}. \tag{4}$$

These are the same τ_g, M and C as in the given model.

D. State-Space Form. M is always invertible, so solving (1) for $\ddot{q}$ gives the state equation:

$$\det M = (2.663 + 2c_2)(1.33) - (1.33 + c_2)^2 = 1.7729 - c_2^2 \ge 0.7729 > 0 \ \ (c_2^2 \le 1), \qquad \dot{x} = f(x,\tau) = \begin{bmatrix} \dot{q} \\ M^{-1}\left(\tau_g + B\tau - C\dot{q}\right) \end{bmatrix}. \tag{5}$$

II. Equilibria

A. Conditions. At rest $\dot{q} = \ddot{q} = 0$, and the input torque is given as $\tau^* = 0$. Then (1) gives $\tau_g(q^*) = 0$:

$$\text{row 2: } \sin(q_1^* + q_2^*) = 0, \qquad \text{row 1: } 1.5\sin q_1^* + \underbrace{\sin(q_1^* + q_2^*)}_{=0} = 0 \ \Rightarrow\ \sin q_1^* = 0.$$

B. Solutions. $q_1^* \in \{0,\pi\}$, $q_1^* + q_2^* \in \{0,\pi\} \Rightarrow q_2^* \in \{0,\pi\}$ (mod 2π): **four equilibria**, all with both links vertical (Table I).

Table I. Equilibria for $\tau^* = 0$

	Posture	(q_1^*, q_2^*)	U [J]	Description
E_1	down–down	$(0,0)$	-24.525	both links hanging
E_2	down–up	$(0,\pi)$	-4.905	link 1 down, link 2 up
E_3	up–up	$(\pi,0)$	$+24.525$	fully upright (target)
E_4	up–down	(π,π)	$+4.905$	link 1 up, link 2 down

III. Linearization

A. Deviation Variables. $\tilde{q} = [q_1 - q_1^*,\ q_2 - q_2^*]^T$, $\tilde{x} = [\tilde{q}; \dot{\tilde{q}}] = x - x^*$, $\tilde{\tau} = \tau - \tau^* = \tau$. First-order Taylor expansion of (5) about $(x^*, 0)$, using $f(x^*, 0) = 0$: $\dot{\tilde{x}} \approx \frac{\partial f}{\partial x}\big|_* \tilde{x} + \frac{\partial f}{\partial \tau}\big|_* \tilde{\tau}$.

B. Jacobians. *Top half:* $\frac{d}{dt}\tilde{q} = \dot{\tilde{q}} \Rightarrow$ rows $[0\ \ I_2]$. *Bottom half:* $h = M^{-1}(\tau_g + B\tau - C\dot{q})$; at $(x^*, 0)$, $\partial(C\dot{q})/\partial\dot{q} = 0$ ($C\dot{q}$ is quadratic in $\dot{q}$, $\dot{q}^* = 0$) and $(\partial M^{-1}/\partial q)(\tau_g + B\tau) = 0$ ($\tau_g(q^*) = 0$, $\tau^* = 0$). Hence

$$\ddot{\tilde{q}} = P\tilde{q} + r\tilde{\tau}, \quad P = M(q^*)^{-1}K(q^*), \quad r = M(q^*)^{-1}B, \qquad K(q^*) = \frac{\partial \tau_g}{\partial q}\Big|_{q^*} = -9.81 \begin{bmatrix} 1.5\cos q_1^* + \cos\varphi^* & \cos\varphi^* \\ \cos\varphi^* & \cos\varphi^* \end{bmatrix}. \tag{6}$$

C. Linear State-Space Model. Stacking both halves, with the joint angles as outputs, $y = \tilde{q} = [q_1 - q_1^*,\ q_2 - q_2^*]^T$:

$$\boxed{\dot{\tilde{x}} = A\tilde{x} + B_\ell \tilde{\tau}, \quad y = C_y\tilde{x} + D\tilde{\tau}, \qquad A = \begin{bmatrix} 0 & I_2 \\ P & 0 \end{bmatrix},\ B_\ell = \begin{bmatrix} 0 \\ r \end{bmatrix},\ C_y = \begin{bmatrix} I_2 & 0 \end{bmatrix},\ D = 0} \tag{7}$$

D. Example: $E_1 = (0,0)$. $\cos q_1^* = \cos\varphi^* = c_2 = 1$:

$$K = \begin{bmatrix} -24.525 & -9.81 \\ -9.81 & -9.81 \end{bmatrix}, \quad M = \begin{bmatrix} 4.663 & 2.33 \\ 2.33 & 1.33 \end{bmatrix}, \quad \det M = 0.7729, \quad M^{-1} = \frac{1}{0.7729}\begin{bmatrix} 1.33 & -2.33 \\ -2.33 & 4.663 \end{bmatrix},$$

$$P_{E1} = M^{-1}K = \begin{bmatrix} -12.6292 & 12.6926 \\ 14.7488 & -29.6119 \end{bmatrix}, \qquad r_{E1} = \begin{bmatrix} -3.0147 \\ 6.0332 \end{bmatrix}.$$

E. All Equilibria. M depends only on q_2^*; K only on the signs of $\cos q_1^*$, $\cos\varphi^*$. Upside down, gravity flips: $K_{E3} = -K_{E1}$ with $M_{E3} = M_{E1}$, so $P_{E3} = -P_{E1}$; likewise $P_{E4} = -P_{E2}$. Results in Table II; with (7) they give (A, B_ℓ, C_y, D) at each equilibrium.

Table II. Linearization results ($\det M = 0.7729$ at all four)

	$(\cos q_1^*, \cos\varphi^*)$	$M(q^*)$	$P = M^{-1}K$	$r = M^{-1}B$
E_1	$(1,1)$	$\left[\begin{smallmatrix} 4.663 & 2.33 \\ 2.33 & 1.33 \end{smallmatrix}\right]$	$\left[\begin{smallmatrix} -12.6292 & 12.6926 \\ 14.7488 & -29.6119 \end{smallmatrix}\right]$	$\left[\begin{smallmatrix} -3.0147 \\ 6.0332 \end{smallmatrix}\right]$
E_2	$(1,-1)$	$\left[\begin{smallmatrix} 0.663 & 0.33 \\ 0.33 & 1.33 \end{smallmatrix}\right]$	$\left[\begin{smallmatrix} -12.6292 & 12.6926 \\ 10.5095 & 4.2266 \end{smallmatrix}\right]$	$\left[\begin{smallmatrix} -0.4270 \\ 0.8578 \end{smallmatrix}\right]$
E_3	$(-1,-1)$	as E_1	$-P_{E1}$	r_{E1}
E_4	$(-1,1)$	as E_2	$-P_{E2}$	r_{E2}

IV. Transfer Functions

A. Laplace Transform. With $\tilde{Q}_i(s) = \mathscr{L}\{\tilde{q}_i(t)\}$, $\tilde{\mathscr{T}}(s) = \mathscr{L}\{\tilde{\tau}(t)\}$, $\tilde{Q} = [\tilde{Q}_1, \tilde{Q}_2]^T$ and zero initial conditions ($\ddot{\tilde{q}} \to s^2\tilde{Q}$):

$$s^2\tilde{Q} = P\tilde{Q} + r\tilde{\mathscr{T}} \ \Rightarrow\ (s^2 I - P)\,\tilde{Q}(s) = r\,\tilde{\mathscr{T}}(s), \qquad (s^2 I - P)^{-1} = \frac{1}{\Delta(s)}\begin{bmatrix} s^2 - P_{22} & P_{12} \\ P_{21} & s^2 - P_{11} \end{bmatrix}, \quad \Delta(s) = s^4 - \mathrm{tr}(P)\,s^2 + \det P.$$

B. General Formula. Multiplying by r and dividing each row by $\tilde{\mathscr{T}}$ ($G_i = \tilde{Q}_i/\tilde{\mathscr{T}}$):

$$\boxed{G_1(s) = \frac{r_1 s^2 + (P_{12}r_2 - P_{22}r_1)}{\Delta(s)}, \qquad G_2(s) = \frac{r_2 s^2 + (P_{21}r_1 - P_{11}r_2)}{\Delta(s)}} \tag{8}$$

Poles: roots of Δ (= eigenvalues of A); with $\lambda = s^2$, $\lambda^2 - \mathrm{tr}(P)\lambda + \det P = 0$, then $s = \pm\sqrt{\lambda}$. *Zeros:* $s^2 = -(P_{12}r_2 - P_{22}r_1)/r_1$ for G_1 and $s^2 = -(P_{21}r_1 - P_{11}r_2)/r_2$ for G_2.

C. Example: E_1. $\operatorname{tr} P = -42.2410$, $\det P = 186.7719 \Rightarrow \lambda = s^2 = -5.018, -37.224 \Rightarrow$ poles $\pm j2.2400$, $\pm j6.1011$; numerator constants $P_{12}r_2 - P_{22}r_1 = -12.6926$, $P_{21}r_1 - P_{11}r_2 = 31.7316 \Rightarrow G_1 = -3.0147\,(s^2+4.2103)/\Delta$, $G_2 = 6.0332\,(s^2+5.2595)/\Delta$ (row E_1 of Table III).

Table III. Transfer functions $G_i(s) = N_i(s)/\Delta(s)$ at the four equilibria

	Denominator $\Delta(s)$ (monic)	$N_1(s)$ $(\tau \to q_1)$	$N_2(s)$ $(\tau \to q_2)$	Poles	Zeros of G_1	Zeros of G_2
E_1	$s^4 + 42.2410\,s^2 + 186.7719$	$-3.0147\,(s^2+4.2103)$	$6.0332\,(s^2+5.2595)$	$\pm j2.2400,\ \pm j6.1011$	$\pm j2.0519$	$\pm j2.2934$
E_2	$s^4 + 8.4025\,s^2 - 186.7719$	$-0.4270\,(s^2-29.7273)$	$0.8578\,(s^2+7.3982)$	$\pm 3.1775,\ \pm j4.3010$	± 5.4523	$\pm j2.7200$
E_3	$s^4 - 42.2410\,s^2 + 186.7719$	$-3.0147\,(s^2-4.2103)$	$6.0332\,(s^2-5.2595)$	$\pm 2.2400,\ \pm 6.1011$	± 2.0519	± 2.2934
E_4	$s^4 - 8.4025\,s^2 - 186.7719$	$-0.4270\,(s^2+29.7273)$	$0.8578\,(s^2-7.3982)$	$\pm 4.3010,\ \pm j3.1775$	$\pm j5.4523$	± 7.2700

D. All Equilibria. Table III. With no friction only even powers of s occur, so poles and zeros come in pairs $\pm p$; since $P_{E3} = -P_{E1}$, each $\pm j\omega$ at E_1 becomes $\pm\omega$ at E_3 (likewise E_2/E_4).

V. Verification

- The model derivation and all values in Tables II and III were verified in MATLAB (agreement to four decimals).
- Nonlinear (5) vs. linear (7), from the same small deviation with $\tau = 0$ (Fig. 2): near E_1 from $\tilde{q}(0) = (0.1, -0.1)$ rad they agree within 1.0° (q_1) and 2.0° (q_2) over 8 s. Near E_3 from $\tilde{q}_1(0) = 10^{-3}$ rad the error is 0.02% when $|\tilde{q}_1| = 1^\circ$.
- With $\tau = 0$, $T + U$ stays constant to 10^{-8} (no friction).
- Phase 1 animation: **watch the video**.

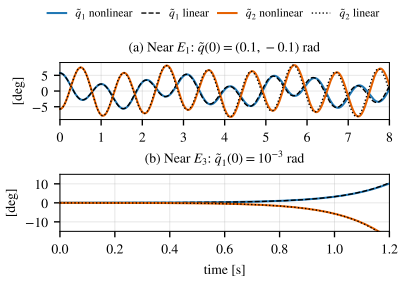


Fig. 2. Nonlinear (colour) vs. linear (black) responses near (a) E_1, (b) E_3.

Phase 2: Stability Analysis of the Acrobot

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Abstract. Using the Phase 1 model (Sections I–V), we classify every equilibrium from the transfer functions and from A, link the result to the shape of U, and confirm it in a PyBullet simulation. At the upright E_3 we prove that static or PD feedback from one joint angle cannot stabilize it, and show that a dynamic observer-based controller can, for small offsets only.

VI. Stability Criteria

A. Definitions. For $\dot{\tilde{x}} = A\tilde{x} + B_\ell \tilde{\tau}$ and $G_j(s) = C_j(sI-A)^{-1}B_\ell = N_j(s)/\Delta(s)$, $C_j = e_j^T$:

$$\begin{array}{lll} \text{asymptotically stable} & \Leftrightarrow\ \mathrm{Re}\,\lambda_i(A) < 0\ \ \forall i, & \text{BIBO stable } (G_j) \quad \Leftrightarrow\ \mathrm{Re}\,p < 0\ \ \forall \text{ poles } p, \\ \text{marginally stable} & \Leftrightarrow\ \mathrm{Re}\,\lambda_i \le 0, \text{ simple on } j\mathbb{R}, & \text{nonlinear (Lyapunov indirect)} \quad : \exists \mathrm{Re}\,\lambda_i > 0 \Rightarrow \text{unstable}. \end{array} \tag{9}$$

B. Poles of G_j = Eigenvalues of A. At every equilibrium

$$\mathrm{rank}\underbrace{[B_\ell\ \ AB_\ell\ \ A^2B_\ell\ \ A^3B_\ell]}_{\mathscr{C}} = 4, \qquad \mathrm{rank}\underbrace{[C_j;\ C_jA;\ C_jA^2;\ C_jA^3]}_{\mathscr{O}_j} = 4 \quad (j=1,2),$$

so each G_j is a minimal realization (no pole–zero cancellation, Table III) and $\{\text{poles of } G_1\} = \{\text{poles of } G_2\} = \{\lambda_i(A)\}$: **the transfer-function and state-space verdicts coincide**.

C. Characteristic Polynomial.

$$\Delta(s) = \det(sI-A) = s^4 + \boxed{0}\,s^3 - \mathrm{tr}(P)\,s^2 + \boxed{0}\,s + \det P. \tag{10}$$

A Hurwitz polynomial has all coefficients > 0; the s^3 and s^1 coefficients are 0 (no friction) $\Rightarrow$ **no equilibrium is asymptotically stable for $\tau = 0$.** With $\lambda = s^2 = \mathrm{eig}(P)$:

$$\lambda^2 - \mathrm{tr}(P)\,\lambda + \det P = 0, \qquad \lambda < 0 \Rightarrow s = \pm j\sqrt{-\lambda}, \qquad \lambda > 0 \Rightarrow s = \pm\sqrt{\lambda}\ \text{(one RHP pole)}. \tag{11}$$

Table IV. Pole pattern from the signs of $\det P$ and $\mathrm{tr}\,P$

$\det P$	$\mathrm{tr}\,P$	λ_1, λ_2	poles	case
> 0	< 0	both < 0	$\pm j\omega_1,\ \pm j\omega_2$	E_1
< 0	any	opposite signs	$\pm\sigma,\ \pm j\omega$	E_2, E_4
> 0	> 0	both > 0	$\pm\sigma_1,\ \pm\sigma_2$	E_3

D. Link to the Potential Energy.

$$K = \frac{\partial \tau_g}{\partial q} = -\nabla^2 U, \qquad P = M^{-1}K \sim M^{-1/2}KM^{-1/2}\ \ (M \succ 0) \overset{\text{Sylvester}}{\Longrightarrow} \boxed{\#\{\text{RHP poles}\} = \#\{\text{negative eigenvalues of } \nabla^2 U\}} \tag{12}$$

Minimum of $U \to 0$ RHP poles, saddle $\to 1$, maximum $\to 2$ [3].

VII. Stability of Each Equilibrium

Table V. Stability of the four equilibria (transfer functions, state matrix and energy agree)

	$(\mathrm{tr}\,P,\ \det P)$	$\lambda = \mathrm{eig}\,P$	poles of $G_1, G_2 = \mathrm{eig}\,A$	$\mathrm{eig}\,\nabla^2 U$	RHP	verdict (linear / nonlinear)
E_1	$(-42.241,\ 186.772)$	$-5.018,\ -37.224$	$\pm j2.2400,\ \pm j6.1011$	4.905, 29.43 (min)	0	marginal / Lyapunov stable
E_2	$(-8.403,\ -186.772)$	$-18.499,\ 10.096$	$\pm 3.1775,\ \pm j4.3010$	-14.715, 9.81 (saddle)	1	unstable / unstable
E_3	$(42.241,\ 186.772)$	$5.018,\ 37.224$	$\pm 2.2400,\ \pm 6.1011$	$-29.43,\ -4.905$ (max)	2	unstable / unstable
E_4	$(8.403,\ -186.772)$	$-10.096,\ 18.499$	$\pm 4.3010,\ \pm j3.1775$	-9.81, 14.715 (saddle)	1	unstable / unstable

A. E_1 ***(down–down).***

- *Transfer function:* simple poles on $j\mathbb{R} \Rightarrow$ not BIBO stable. E.g. $\tau = \sin(2.24t)$: $\tilde{Q}_j(s)$ gets a double pole at $\pm j2.24 \Rightarrow \tilde{q}_j \sim t\sin(2.24t)$.
- *State space:* $\tilde{x}(t) = \sum_{k=1}^{2} \left(a_k \cos\omega_k t + b_k \sin\omega_k t\right) v_k$, $\omega_{1,2} = 2.24, 6.10 \Rightarrow$ bounded, non-decaying: **marginally stable**.
- *Nonlinear (linearization inconclusive):* Lyapunov function [5]

$$V(x) = \tfrac{1}{2}\dot{q}^T M(q)\dot{q} + U(q) - U(q^*), \qquad \nabla^2 U(q^*) \succ 0 \Rightarrow V \succ 0 \text{ near } E_1, \qquad \dot{V} = \dot{q}^T B\tau = 0. \tag{13}$$

$\Rightarrow$ **stable in the sense of Lyapunov, not asymptotically** (V is constant: the swing never decays).

B. E_2***,*** E_4 ***(one link up).*** $\det P = -186.772 < 0 \Rightarrow \lambda_1\lambda_2 < 0 \Rightarrow$ one real pole in the RHP ($+3.1775$ at E_2, $+4.3010$ at E_4) $\Rightarrow$ BIBO unstable, A unstable, nonlinear unstable by (9). U has a saddle.

C. E_3 ***(up–up).*** $P_{E3} = -P_{E1} \Rightarrow \lambda(P_{E3}) = -\lambda(P_{E1}) = \{5.018, 37.224\} \Rightarrow s = \pm 2.2400,\ \pm 6.1011$: **two RHP poles**, unstable in every sense. U has a strict maximum.

VIII. Simulation Evidence

A. Simulator. Nonlinear model in **PyBullet** [6] (uniform rods, no damping, step $h = 0.25$ ms), run alongside the linear model (7) from the same initial condition. Check against RK4 of (1) with $q(0) = (0.8, -0.5)$ rad, $\tau = 3\sin 2t$: $\max_{t \le 5\,\mathrm{s}} |q_{\mathrm{PyBullet}} - q_{\mathrm{RK4}}| = 0.52°$.

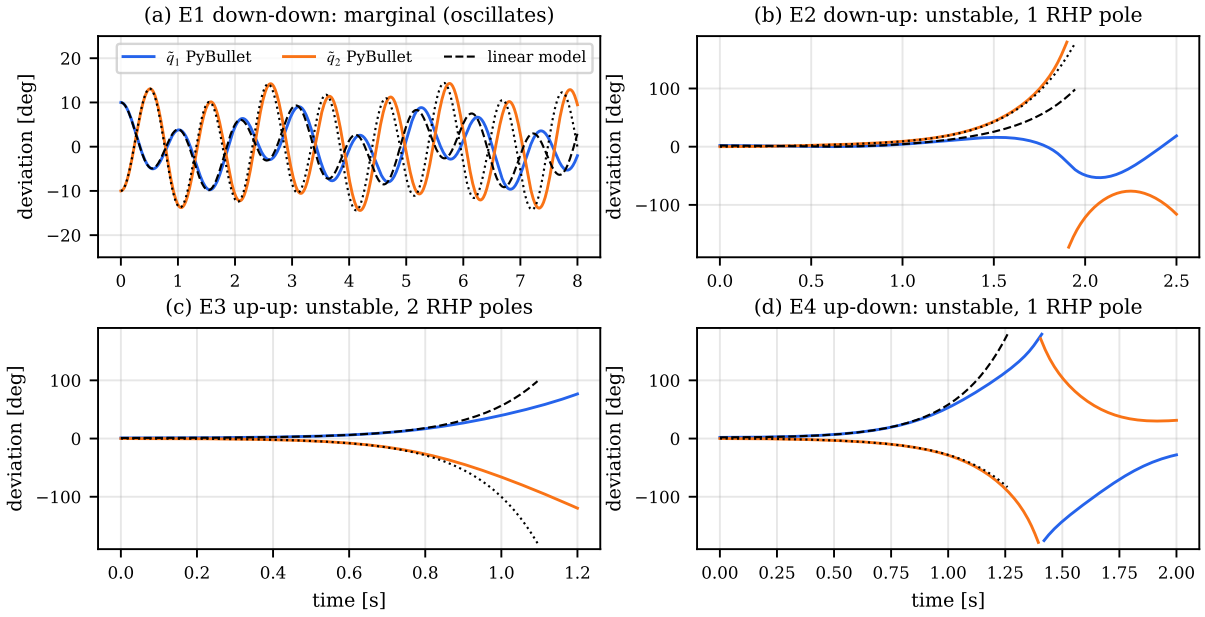


Fig. 3. Free response ($\tau = 0$) near each equilibrium: PyBullet (colour) vs. linear model (black). Initial offsets: (a) $(10°, -10°)$; (b) $2°$; (c) $1°$; (d) $2°$ on q_1.

B. Prediction vs. Measurement. Near an unstable equilibrium $\|\tilde{q}(t)\| \approx c\,e^{p_{\max} t}$, so the measured rate is $\hat{\sigma} = \mathrm{d}\ln\|\tilde{q}\|/\mathrm{d}t$ (least-squares fit for $0.2° < \|\tilde{q}\| < 15°$).

Table VI. Linear-model predictions vs. PyBullet measurements

	quantity	predicted (linear)	measured (PyBullet)	error
E_1	oscillation frequencies (60 s run)	2.240, 6.101 rad/s	2.239, 5.997 rad/s	within 1 bin ($2\pi/60 = 0.105$)
E_1	energy change over 60 s	0	1.2×10^{-3} J	integration error
E_2	growth rate $\hat{\sigma}$; time to 30° from 0.05°	3.1775 s^{-1}	3.171 s^{-1}; 2.52 s	0.22%
E_3	growth rate $\hat{\sigma}$; time to 30° from 0.01°	6.1011 s^{-1}	6.088 s^{-1}; 1.55 s	0.21%
E_4	growth rate $\hat{\sigma}$; time to 30° from 0.05°	4.3010 s^{-1}	4.294 s^{-1}; 1.69 s	0.15%

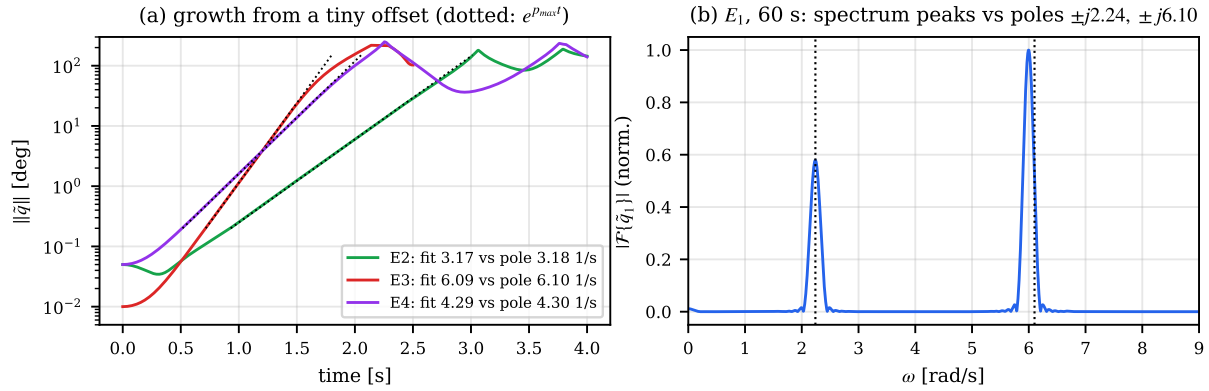


Fig. 4. (a) $\|\tilde{q}\|$ from a tiny offset; dotted slope $= p_{\max}$. (b) Spectrum of $\tilde{q}_1$ near E_1; dotted lines are the poles.

IX. Single-Angle Feedback at the Upright E_3

A. Loop. Measure one angle, $y = \tilde{q}_j$, and apply $\tau = -C(s)\,y$ with $C = N_c/D_c$:

$$1 + C(s)G_j(s) = 0 \;\Leftrightarrow\; \Delta(s)D_c(s) + N_j(s)N_c(s) = 0. \tag{14}$$

At E_3 (Table III), writing $\Delta = s^4 + a_2 s^2 + a_0$ and $N_j = b_2 s^2 + b_0$:

Table VII. Plant data at E_3 ($a_2 = -42.241$, $a_0 = 186.772$)

measured	b_2	b_0	b_0/b_2	zeros $\pm z$	RHP poles
q_1	-3.0147	12.6926	-4.2103	± 2.0519	2.2400, 6.1011
q_2	6.0332	-31.7316	-5.2595	± 2.2934	2.2400, 6.1011

B. Static and PD Feedback: Impossible. With $C(s) = K_p + K_d s$, (14) becomes

$$s^4 + (K_d b_2)\,s^3 + (a_2 + K_p b_2)\,s^2 + (K_d b_0)\,s + (a_0 + K_p b_0) = 0. \tag{15}$$

Proof: Hurwitz $\Rightarrow K_d b_2 > 0$ and $K_d b_0 > 0 \Rightarrow b_0/b_2 > 0$. But $b_0/b_2 = -z^2 < 0$ for both angles (Table VII) $\Rightarrow$ contradiction. ■

- $K_d = 0$: (15) is even, $p(s) = p(-s) \Rightarrow$ roots symmetric about $j\mathbb{R} \Rightarrow$ never Hurwitz.
- Root locus: gain $\to 0$, roots $\to \{\pm 2.24, \pm 6.10\}$; gain $\to \infty$, roots $\to \{-K_p/K_d,\ \pm z\} \ni +z \Rightarrow$ a pole always remains in the RHP. The cause is the **real RHP zero** $+z$.
- Grid $K_p \in [-400, 400]$, $K_d \in [-150, 150]$: $\min \max \mathrm{Re}\,\lambda = 0.96\ \mathrm{s}^{-1}$ (q_1, at $K_p = -15$, $K_d = 0$) and $2.29\ \mathrm{s}^{-1}$ (q_2, $\to z = 2.2934$ at high gain). PyBullet with these gains: $1^\circ \to 30^\circ$ in 1.0 s (q_1) and 1.8 s (q_2).

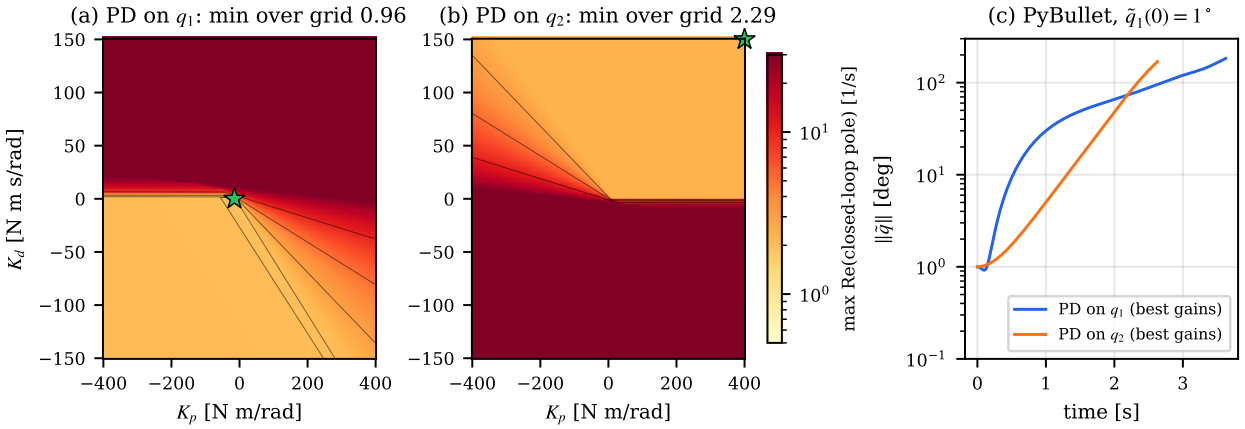


Fig. 5. PD from one angle at E_3. (a, b) max Re of the closed-loop poles over the gain plane: positive everywhere (star: best gains). (c) PyBullet with the best gains.

C. Dynamic Output Feedback: Possible but Fragile. $\mathcal{O}_j$ has rank 4, so an observer reconstructs $\tilde{x}$ from $y = \tilde{q}_j$ alone:

$$\dot{\hat{x}} = A\hat{x} + B_\ell \tau + L(y - C_j\hat{x}), \qquad \tau = -K\hat{x}, \qquad \lambda_{\text{cl}} = \lambda(A - B_\ell K) \cup \lambda(A - LC_j) \quad \text{(separation)}. \tag{16}$$

K from LQR with $Q = \text{diag}(10, 10, 1, 1)$, $R = 1$: $K = [-266.1, -107.4, -115.0, -54.3]$, $\lambda(A - B_\ell K) = \{-10.09, -4.25, -2.24 \pm 0.04j\}$. L from the dual LQR problem. Two limits hold for *any* controller:

$$\text{PIP [4]: stable } C(s) \text{ exists} \Leftrightarrow \#\{\text{real RHP poles in } (z, \infty)\} \text{ even}; \qquad \|S\|_\infty \geq \prod_i \frac{|z + p_i|}{|z - p_i|} \text{ [3]}. \tag{17}$$

z sits just below the RHP pole 2.24, so $|z - p_1|$ is small and $\|S\|_\infty$ is huge: the loop is very sensitive, and saturation at 20 N m breaks it for offsets beyond the last column of Table VIII.

Table VIII. Single-angle observer + LQR at E_3: design, fundamental limits and PyBullet results ($|\tau| \leq 20$ N m)

measured	observer poles $\lambda(A - LC_j)$	poles in (z, ∞)	stable controller?	$\|S\|_\infty \geq$	largest offset held: $\tilde{q}_1(0)$, $\tilde{q}_2(0)$
q_1	$-6.34, -5.88, -2.67, -1.90$	2.24, 6.10 (even)	yes	45.9	1.42°, 1.66°
q_2	$-6.56, -5.67, -2.49, -2.03$	6.10 (odd)	**no** (must be unstable)	187	0.23°, 0.97°
full state	–	–	–	–	4.64°, 10.06°

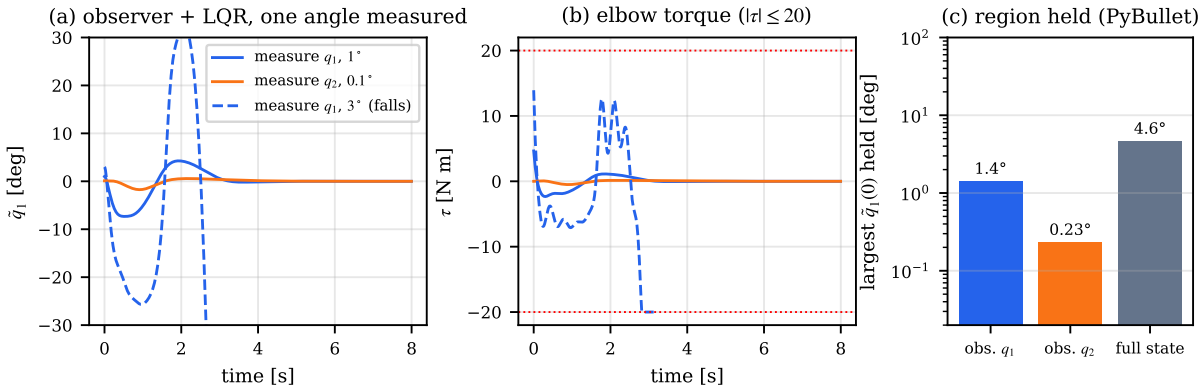


Fig. 6. Observer + LQR at E_3 in PyBullet using one angle: (a) $\tilde{q}_1$; (b) torque; (c) largest initial offset still stabilized.

D. Answer.

- **Static / PD feedback from one angle cannot stabilize** E_3 (either angle): the real RHP zero gives $b_0/b_2 < 0$, which contradicts the Hurwitz conditions of (15).
- **A dynamic controller can**, since (A, C_j) is observable for $j = 1, 2$. Via q_1 a stable controller suffices; via q_2 it must be unstable (PIP).
- **It is fragile**: $\|S\|_\infty \geq 46$ (q_1) or 187 (q_2), and with $|\tau| \leq 20$ N m it holds only $\approx 1.4^\circ$ (q_1) or $\approx 0.2^\circ$ (q_2), vs. 4.6° with full state. Full-state feedback is the route for Phase 3.

X. Simulation Tools

- **Interactive PyBullet simulator:** `phase2/acrobot_phase2_sim.py`, instructions in `phase2/README.md`. Live dashboard: choose equilibrium, initial condition (slider, typed value or drag), and controller (open loop, single-angle PD, single-angle observer + LQR); shows the nonlinear arm with the linear model, $\tilde{q}(t)$, $\tau(t)$ and the closed-loop poles.
- **Reproducibility:** every figure and number above comes from `code/make_phase2_results.py` in the Phase 2 deliverable folder.
- **Phase 2 animation** (Manim, 2K): **watch the video**.

Phase 3: Control Around the Equilibria

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Control Systems Project · Phase 3 Report · November 5, 2026

Abstract. We stabilize all four equilibria: the upright E_3 and E_2, E_4 with full-state LQR, and E_1 with a PD controller on the elbow angle alone. Each choice is justified by a weight sweep or a Lyapunov argument. We then measure how far from each equilibrium the controller still recovers with $|\tau| \le 20\,\mathrm{N\,m}$, both analytically and with region-of-attraction maps of the nonlinear model.

XI. Design Approach

A. Why Full-State Feedback. Phase 2 showed that one measured angle is not enough at E_3: PD is impossible and an observer holds only $\approx 1^\circ$. All four states are therefore fed back, which is possible because the system is controllable at every equilibrium ($\operatorname{rank}\mathscr{C} = 4$, Sec. VI-B). The motor is limited, so the applied torque is

$$\tau = \mathrm{sat}\big(-K\tilde{x}\big) = \max\big(-20,\ \min(20, -K\tilde{x})\big), \qquad \tilde{x} = [\tilde{q}_1,\ \tilde{q}_2,\ \dot{q}_1,\ \dot{q}_2]^T. \tag{18}$$

B. LQR. K minimizes $J = \int_0^\infty \big(\tilde{x}^T Q\tilde{x} + R\,\tau^2\big)\,dt$ for the linear model (7):

$$A^T S + SA - SB_\ell R^{-1} B_\ell^T S + Q = 0, \qquad K = R^{-1} B_\ell^T S. \tag{19}$$

- *Why LQR:* the closed loop is stable for any $Q \succeq 0$, $R > 0$ with (A, B_ℓ) controllable, and it keeps a phase margin $\ge 60^\circ$ and gain margin $[\frac{1}{2}, \infty)$ [7]. One scalar (R) trades speed against torque.
- *Weights:* $Q = I$ weighs 1 rad of angle error like 1 rad/s of rate error. R is chosen by the sweep below.

C. Choice of R. For each R we compute K and simulate the nonlinear saturated loop from a grid of initial offsets (Sec. XIII-B). Fig. 7:

- E_3: the recovered area is almost flat (16–20% of a $\pm 60^\circ$ grid for $10^{-3} \le R \le 1$), while the fastest pole goes from -10 to $-214\ \mathrm{s}^{-1}$. The limit comes from $\tau_{\max}$, not from tuning. $R = 0.03$ is the knee: 93% of the best area with a fastest pole of -40 instead of -214.
- E_2: the region collapses for $R > 0.3$ (99% $\to$ 90% $\to$ 20% at $R = 0.3, 0.56, 1$) $\Rightarrow R = 0.1$. E_4: the area peaks at $R \approx 0.1$–0.18 (22–24%) and drops on both sides $\Rightarrow R = 0.1$.

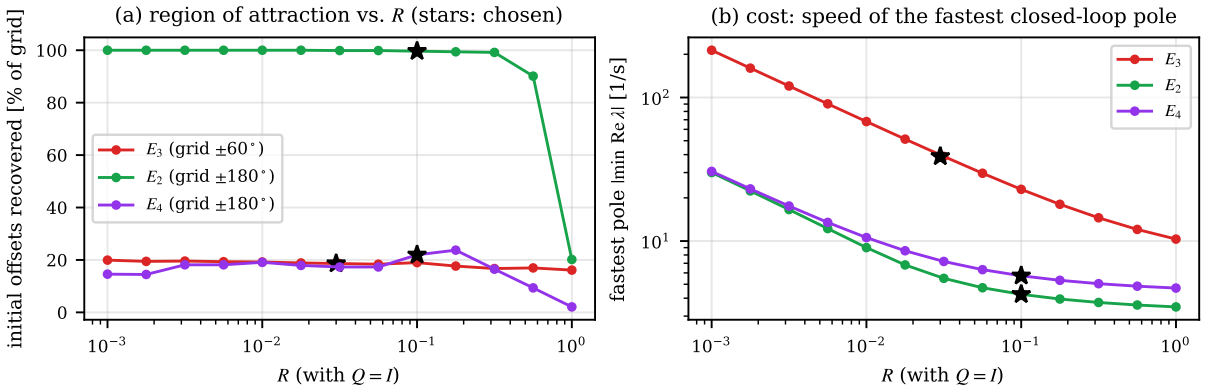


Fig. 7. Weight selection ($Q = I$). (a) Share of initial offsets recovered by the nonlinear saturated loop. (b) Fastest closed-loop pole, a measure of bandwidth and torque demand. Stars: chosen R.

D. Comparison: Pole Placement at E_3. Pole placement (`place`) at $\{-3, -4, -5, -6\}$ gives $K = [-322.9, -134.6, -138.1, -66.0]$. It settles faster for small offsets (1.3 s vs 4.1 s from 2°) with less torque (11.3 vs 17.6 N m), but recovers a smaller region (20% vs 26% of $\pm 40^\circ$; 5° vs 6° along q_1). LQR is kept for E_3.

E. E_1***: PD on the Elbow Angle Only.*** At E_1 the zeros of G_2 are imaginary ($\pm j2.29$, Table III), so the sign conflict of (15) does not occur and a single angle suffices:

$$\tau = -K_p\tilde{q}_2 - K_d\dot{q}_2, \qquad K_p = -5,\ K_d = 0.4 \quad \big(\text{best decay rate on a } (K_p, K_d) \text{ grid: } \max \mathrm{Re}\,\lambda = -0.52\big). \tag{20}$$

Proof of stability (energy shaping, LaSalle [5]): with $E = T + U$, $\dot{E} = \dot{q}^T B\tau = \dot{q}_2\tau$, so

$$V_c = E - E^* + \tfrac{1}{2}K_p\tilde{q}_2^2, \qquad \dot{V}_c = -K_d\,\dot{q}_2^2 \le 0, \qquad \nabla^2\big(U + \tfrac{1}{2}K_p\tilde{q}_2^2\big)\big|_{E_1} :\ \lambda = 0.760,\ 28.575 > 0. \tag{21}$$

V_c has a strict minimum at E_1 and never increases (while $|\tau| < 20$), and $\dot{q}_2 \equiv 0$ forces E_1, so E_1 is asymptotically stable. Pure damping ($K_p = 0$) also works but decays 19× more slowly ($\max \mathrm{Re}\,\lambda = -0.027$): the slow mode swings both links together and barely moves the elbow.

XII. Controllers

Table IX. Phase 3 controllers ($\tau = \mathrm{sat}(-K\tilde{x})$) and their linearized closed-loop poles

controller	design	$K = [k_{q_1},\ k_{q_2},\ k_{\dot{q}_1},\ k_{\dot{q}_2}]$	closed-loop poles
E_3 LQR	$Q = I, R = 0.03$	$[-503.0,\ -203.9,\ -218.3,\ -101.5]$	$-39.86,\ -2.25 \pm 0.10j,\ -1.34$
E_3 pole placement	$\{-3, -4, -5, -6\}$	$[-322.9,\ -134.6,\ -138.1,\ -66.0]$	$-6,\ -5,\ -4,\ -3$
E_2 LQR	$Q = I, R = 0.1$	$[6.60,\ 46.49,\ 5.23,\ 13.24]$	$-4.26,\ -2.40,\ -1.24 \pm 4.16j$
E_4 LQR	$Q = I, R = 0.1$	$[-49.31,\ 38.91,\ -13.19,\ 6.10]$	$-5.73,\ -3.21,\ -0.96 \pm 3.08j$
E_1 elbow PD	$K_p = -5, K_d = 0.4$	$[0,\ -5,\ 0,\ 0.4]$	$-0.69 \pm 2.31j,\ -0.52 \pm 2.14j$

All poles are in the LHP, so every controller stabilizes its equilibrium locally. The E_3 gains are about 10× larger than at E_2/E_4: the upright has two RHP poles, up to $+6.10$.

XIII. Limits with $\tau_{\max} = 20\,\mathrm{N\,m}$

A. Guaranteed Region (Analytic). Inside the level set $\Omega_c = \{\tilde{x}^T S\tilde{x} \le c\}$ the command never saturates when

$$\max_{\tilde{x}\in\Omega_c} |K\tilde{x}| = \sqrt{cKS^{-1}K^T} \le \tau_{\max} \ \Rightarrow\ c^* = \frac{\tau_{\max}^2}{KS^{-1}K^T}, \qquad \dot{V} = -\tilde{x}^T\big(Q + K^T RK\big)\tilde{x} < 0 \ \text{ on } \Omega_{c^*}. \tag{22}$$

So the linear loop started in Ω_{c^*} stays unsaturated and converges. At rest ($\dot{q} = 0$) this guarantees $|\tilde{q}_1| \le \sqrt{c^*/S_{11}}$ or $|\tilde{q}_2| \le \sqrt{c^*/S_{22}}$: $0.72°/1.66°$ (E_3), $45.6°/19.9°$ (E_2), $20.0°/23.1°$ (E_4). The bound is conservative because the loop usually survives short saturation.

B. Simulated Region of Attraction. The real limits come from simulating the nonlinear saturated loop (18) from a grid of offsets $(\tilde{q}_1(0), \tilde{q}_2(0))$ at rest. A start counts as recovered if after 10 s (30 s for E_1) $|\tilde{q}_i| < 1°$ and $|\dot{q}_i| < 0.05$ rad/s, modulo 2π. The grids are 81×81 ($\pm 40°$) at E_3 and 73×73 ($\pm 180°$) elsewhere. They use an RK4 integration ($h = 2$ ms) of (1), which matches PyBullet on 100% of 45 random grid points and 96% of 45 points on the boundary. The region is not one solid piece: beyond its main part there are thin stripes of starts that swing away and are caught again. The axis limits quoted below are therefore the unbroken range out from q^*.

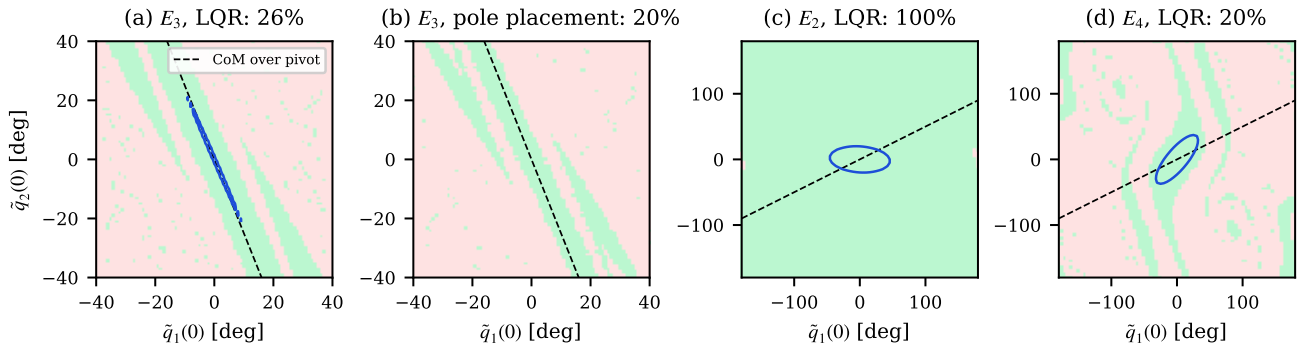


Fig. 8. Region of attraction (green: recovered) with $|\tau| \le 20\,\mathrm{N\,m}$, starting at rest. Blue ellipse: guaranteed region (22). Dashed: offsets with the centre of mass over the pivot.

Table X. How far from the equilibrium each controller still works ($|\tau| \le 20\,\mathrm{N\,m}$, nonlinear model)

controller	recovered (grid)	$\lvert\tilde{q}_1(0)\rvert$ alone	$\lvert\tilde{q}_2(0)\rvert$ alone	along CoM line	guaranteed $\tilde{q}_1$ / $\tilde{q}_2$	shoulder kick $\lvert\dot{q}_1(0)\rvert$	settling (2%)	peak $\lvert\tau\rvert$
E_3 LQR	26% ($\pm 40°$)	6°	13°	$\ge 43°$ (grid edge)	0.72° / 1.66°	0.20 rad/s	4.1 s (2°)	17.6 (2°)
E_3 pole pl.	20% ($\pm 40°$)	5°	9°	$\ge 43°$ (grid edge)	–	0.15 rad/s	1.3 s (2°)	11.3 (2°)
E_2 LQR	99.7% ($\pm 180°$)	175°	180°	full grid	45.6° / 19.9°	≥ 8 rad/s	3.0 s	20 (sat. 3%)
E_4 LQR	20.5% ($\pm 180°$)	35°	55°	47°	20.0° / 23.1°	2.1 rad/s	3.6 s	4.5
E_1 elbow PD	77.8% ($\pm 180°$)	40°	115°	109°	–	2.15 rad/s	12.8 s	5.3

Settling and peak torque: E_3 from $\tilde{q}_1(0) = 2°$; E_2 from $(60°, 90°)$; E_4 from $(20°, 30°)$; E_1 from $(60°, -60°)$ (PyBullet).

C. Why the Upright Region Is a Thin Band. Linearizing τ_g at E_3 ($K_{E3} = -K_{E1}$, Table II):

$$\tau_{g,1} = 9.81\,(2.5\,\tilde{q}_1 + \tilde{q}_2), \qquad \tau_{g,2} = 9.81\,(\tilde{q}_1 + \tilde{q}_2). \tag{23}$$

Gravity pushes the *passive* shoulder with a torque $\propto 2.5\tilde{q}_1 + \tilde{q}_2$, the horizontal offset of the total centre of mass. The motor cannot act on the shoulder directly; it must cancel this torque through the coupling M_{12}, which costs elbow torque. Along $\tilde{q}_2 = -2.5\,\tilde{q}_1$ (CoM over the pivot) there is no shoulder torque, and the region stretches to the grid edge. Across that line every degree of CoM offset costs torque, so $20\,\mathrm{N\,m}$ is used up after 6° in q_1: the 2° response already peaks at $17.6\,\mathrm{N\,m}$. A shoulder kick of only 0.2 rad/s is enough to break it.

XIV. Time Responses (PyBullet)

- E_3 LQR (Fig. 9): held from $\tilde{q}_1(0) = 5.5°$ (torque briefly saturated, 0.13 s in total), falls from 7.5° (bang–bang torque, arm swings away). Held from $(-10°, 25°)$, a large but CoM-balanced offset, with peak torque only $3.3\,\mathrm{N\,m}$.
- E_2, E_4, E_1 (Fig. 10): each converges from a large offset. For E_1, V_c in (21) decreases monotonically, as the proof requires.

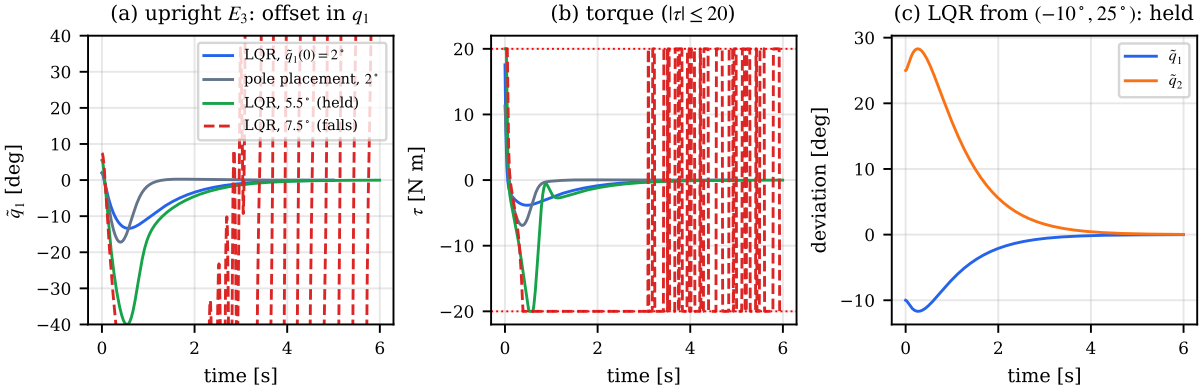


Fig. 9. Upright E_3 in PyBullet. (a) $\tilde{q}_1$ for LQR and pole placement from several offsets in q_1; (b) torque; (c) a CoM-balanced offset.

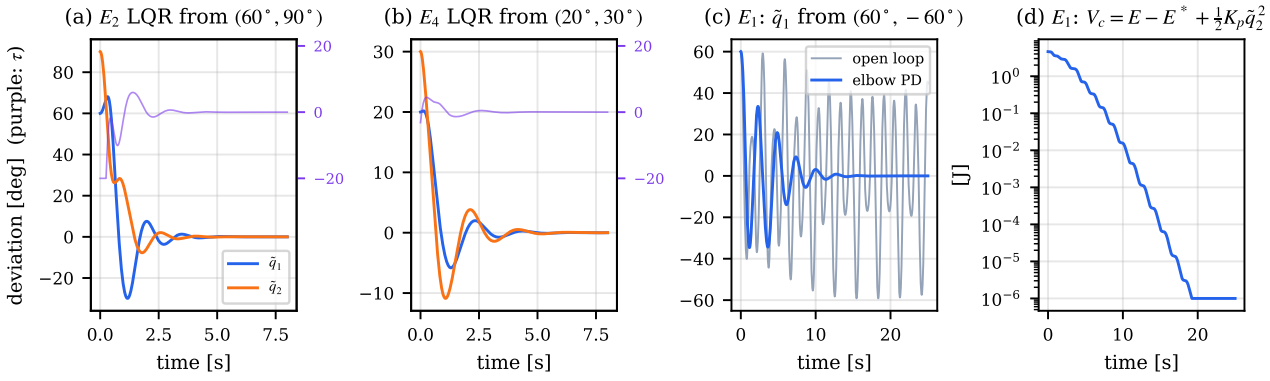


Fig. 10. (a) E_2 and (b) E_4 with LQR (purple: torque); (c) E_1 with elbow PD vs. open loop; (d) V_c of (21) along the E_1 run.

A. Answer.

- **Upright E_3:** full-state LQR ($Q = I$, $R = 0.03$). It recovers from 6° in q_1 or 13° in q_2, and from $\geq 43^\circ$ along the CoM-over-pivot line; 0.72° is guaranteed analytically. The limit is $\tau_{\max}$, not the weights.
- **E_2:** LQR ($R = 0.1$) recovers from almost any rest position (99.7% of $\pm 180^\circ$). **E_4:** LQR ($R = 0.1$), $35^\circ/55^\circ$ along each axis.
- **E_1:** PD on the elbow angle alone, proven asymptotically stable by energy shaping; 78% of $\pm 180^\circ$ recovered.

XV. Simulation and Code (Phase 3)

- **Interactive simulator:** `phase3/acrobot_phase3_sim.py`, instructions in `phase3/README.md`. It is the Phase 2 dashboard plus the Phase 3 controllers (full-state LQR, pole placement, elbow PD).
- **MATLAB:** `code/matlab/Acrobot_Phase3_Main.m` designs every controller (`lqr`, `place`), prints Tables IX and the guaranteed regions, runs the nonlinear loops with `ode45` and maps the regions of attraction. Its results match the Python ones.
- **Reproducibility:** every figure and number in Phase 3 comes from `code/make_phase3_results.py`.
- **Phase 3 animation** (Manim, 2K): **watch the video**.

References. [1] M. W. Spong, "The swing up control problem for the Acrobot," *IEEE Control Syst. Mag.*, vol. 15, no. 1, pp. 49–55, 1995. [2] R. Tedrake, *Underactuated Robotics*, MIT, 2024. [Online]. Available: https://underactuated.csail.mit.edu [3] S. Skogestad and I. Postlethwaite, *Multivariable Feedback Control*, 2nd ed., Wiley, 2005. [4] D. C. Youla, J. J. Bongiorno and C. N. Lu, "Single-loop feedback-stabilization of linear multivariable dynamical plants," *Automatica*, vol. 10, pp. 159–173, 1974. [5] H. K. Khalil, *Nonlinear Systems*, 3rd ed., Prentice Hall, 2002. [6] E. Coumans and Y. Bai, "PyBullet, a Python module for physics simulation," 2016–2024. [Online]. Available: https://pybullet.org [7] B. D. O. Anderson and J. B. Moore, *Optimal Control: Linear Quadratic Methods*, Prentice Hall, 1990.